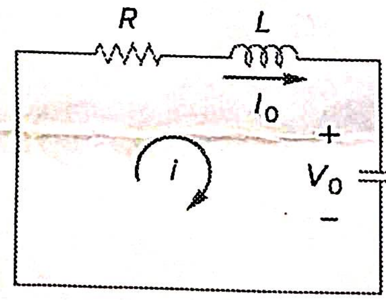
Indian Institute of Technology Jammu
Department of Electrical Engineering
Class Test- 2
Basic Electrical and Electronics Engineering EE-101

Time: 50 min

Note:
- **No extra sheets will be provided.**
- **Kindly write fairly and avoid unnecessary explanation.**

Q1:	A silicon diode conducts a forward current I_F=2.0 mA at $V_{F,25}$=0.7 V when the temperature is 25°C. At the same time, the diode reverse leakage current at −10 V is $I_{R,25}$=20 nA AT 25° C. Calculate the following for temperature 75°C: A. Forward voltage $V_{F,75}$ at the same forward current. B. Forward power $P_{F,25}$ and $P_{F,75}$ and the percentage change in forward power. C. Reverse leakage current $I_{R,75}$ at −10 V. D. Reverse leakage power at −10 V for both temperatures, and comment on its significance compared to forward power. E. Draw the characteristics for both the temperature and for both, reverse and forward bias conditions.	**03**
Q2:	Consider a uniformly doped silicon pn junction at T = 300 K. At zero bias, 25 percent of the total space charge region is in the n-region. The built-in potential barrier is V_{bi} = 0.710 V. Determine (i) N_A, (ii) N_D, (iii) x_n, (iv) x_p. $N_i \Rightarrow 1.5\times10^{10}$	**03**
Q3:	In the figure, R = 40 Ω, L = 4 H, and C = 1/4 F. A. If R=0 Ω, what is the resonant frequency of the circuit? draw V_o with respect to time. B. Calculate the characteristic roots of the circuit. Is the natural response overdamped, underdamped, or critically damped? C. If the inductor values increased by two times, what should be the value for C so that the nature of the circuit is critically damped?  	**04**